

ADITYA KULKARNI

(281) 876-7066 | adityakulkarnius@gmail.com | linkedin.com/in/adityakulkarni244 | github.com/Aditya-k24 | kulkarniaditya.com

SUMMARY

Full-stack Software Engineer (MS CS, NC State, May 2026) who ships features end-to-end, from **PostgreSQL** schema to **React.js** UI. Builds typed **REST APIs**, real-time systems, and cloud-native services in **TypeScript**, **Node.js**, and **Java**; strong in **Docker**, **CI/CD**, **AWS**, and **GCP**.

EXPERIENCE

NC State University

Graduate Research Assistant

Apr 2025 – Present
Raleigh, NC

- Designed a cloud-based genomic platform (**JBrowse 2** on **Google Cloud Run**) enabling **100+ researchers** worldwide to interactively explore **500GB** of strawberry and grape genome datasets
- Built interactive data views and **ML** pipelines, improving model accuracy by **12%** via hyperparameter tuning and time-series analysis

Isomer AI

Product Engineer Intern

May 2025 – May 2026
Remote

- Shipped **10+ features** end-to-end (**PostgreSQL** schema to **React.js** UI) on a per-tenant insurance SaaS; owned **Auth0** RBAC, **Redis** caching, and **Prisma** data modeling
- Built typed **REST** and **MCP** service layers in **Node.js**, **Express.js**, and **TypeScript** backing multi-step LLM agent pipelines and document workflows
- Engineered **Temporal.io** durable workflows with versioned releases and canary splits; shipped via **Docker** + **GitHub Actions** CI/CD with **Sentry** observability across per-tenant stacks

Mphasis (HP Company)

Software Developer Intern

Jun 2024 – Aug 2024
Bangalore, India

- Built **5+ AI workflows**, including a health-claim automation tool using **OCR**, **NLP**, and **ML** classification, that improved automated claim accuracy by **25%**
- Automated deployments on **AWS** and **GCP** with **Grafana** monitoring dashboards, and supported model validation and red-teaming to enhance production reliability

Capgemini

AI/ML Intern

Jun 2023 – Aug 2023
Mumbai, India

- Led the SDLC for a support-ticket automation system, cutting manual processing time **15%**; built **Hadoop**, **Spark**, and **Kafka** pipelines that reduced processing latency **20%**

PROJECTS

PulseRank | Real-Time Leaderboard & Activity Feed

2026

- Architected an event-driven leaderboard: **Redis** sorted sets for **sub-millisecond** reads, a **MySQL** outbox for durable writes, **MongoDB** TTL feed projections, and live updates over Server-Sent Events
- Built end-to-end in **TypeScript**, **NestJS**, and **Next.js**; verified multi-store consistency with **Testcontainers**, instrumented with **OpenTelemetry**, and load-tested with **k6**

InventoryReservationService | Transactional REST API

2026

- Engineered a **Java 21** + **Spring Boot** inventory API preventing overselling under high concurrency via **PostgreSQL** row-level locking (**SELECT FOR UPDATE**) and idempotent reservation endpoints
- Hardened with **Flyway** migrations, **Spring Security** RBAC, RFC 9457 error responses, **Testcontainers** concurrency tests, and a Dockerized **GitHub Actions** CI pipeline

AlgoPulse | Event-Driven Streaming Platform

2025

- Diagnosed an API-polling bottleneck and re-architected to **Supabase Edge Functions** streaming reasoning loops to the client at **2,000 async req/sec**, backed by **Temporal.io** workflows and **Kafka**

CortexQ | Kubernetes Service Platform

2026

- Built a **Kubernetes**-native platform: **FastAPI** router with circuit breaker and load-aware routing, a **KEDA** scale-to-zero autoscaler, and **ArgoCD** GitOps; 88 pytest + 500 VU k6 load tests

EDUCATION

NC State University

M.S. Computer Science

Aug 2024 – May 2026
Raleigh, NC

- Relevant Coursework:** Software Engineering, Object-Oriented Development, Algorithms, Operating Systems, Data Science, Machine Learning

Mumbai University

B.Tech Computer Science, CGPA 9.32/10

Aug 2020 – May 2024
Mumbai, India

- Relevant Coursework:** Data Structures & Algorithms, Big Data Infrastructure, Data Warehousing, Blockchain, Software Engineering

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, Ruby, SQL

Frontend: React.js, Next.js, Vite, HTML/CSS, Server-Sent Events

Backend & APIs: Node.js, Express.js, NestJS, Spring Boot, FastAPI, REST, Prisma

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, Elasticsearch

DevOps & Tools: Docker, Kubernetes, GitHub Actions, AWS, GCP, Temporal.io, Kafka, Playwright, Testcontainers, Grafana

Achievements: HackNC24 Top 10; Sanjeevani Top 1% of 1100+ teams; IEEE-published; 110+ RSpec tests on Expertiza Rails OSS